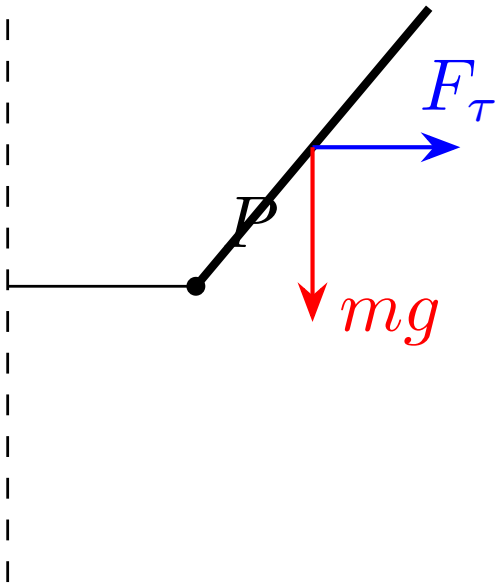
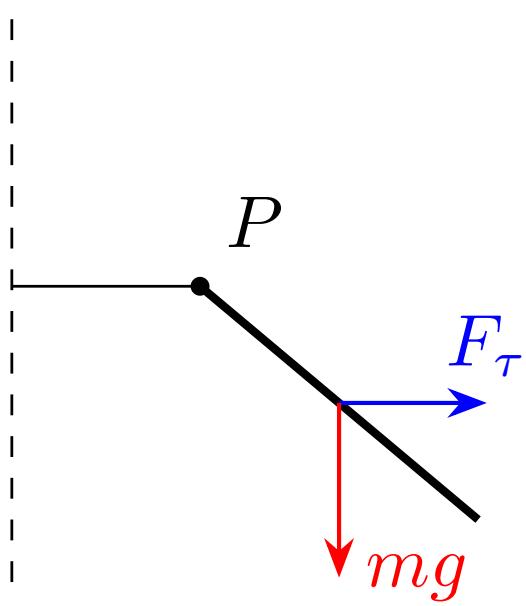
I

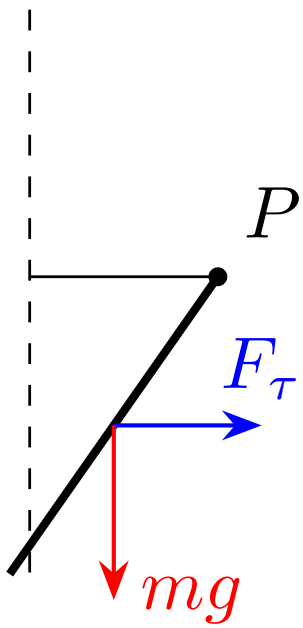


II

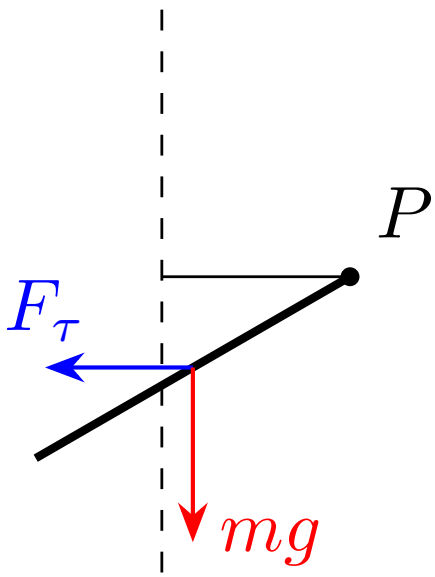
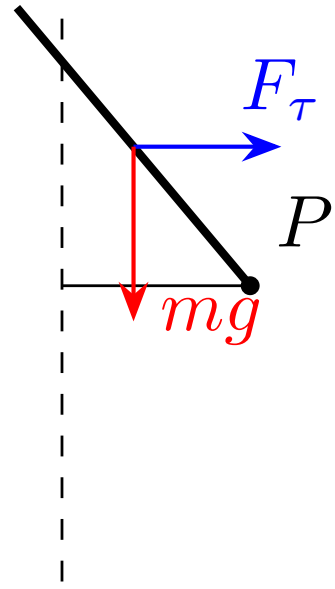


F_τ 仅等效力矩
作用点取质心
非离心力真实合力

III 短杆



III 长杆



IV